SHIYI HE

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UX Researcher | LLM Agents & Evaluation | Human-AI Interaction

Computing Ph.D. student translating behavioral evidence into design recommendations for AI-assisted workflows. Combines quantitative usability studies, practitioner interviews, and human-centered model evaluation.

EDUCATION

University of Utah | Ph.D. Computing — Aug 2023 - Present
University of Wisconsin-Madison | M.S. Information — May 2023
East China Normal University | B.S. Information Management and Systems — Jun 2021

RESEARCH METHODS AND TOOLS

Methods: Mixed-methods research, usability testing, practitioner interviews, think-aloud studies, qualitative analysis.
Synthesis and evaluation: Task performance, workload, quality rubrics, human-AI agreement, design recommendations.
LLM & agent evaluation: Benchmark design, evaluation harnesses, rubric-based LLM-as-a-judge, multi-step agentic judging, human evaluation design, human-judge agreement analysis.
Programming & Tools: Python, SQL, pandas/NumPy, ReVISit, JupyterLite; LLM APIs, LangChain, LangGraph, PyTorch, Hugging Face (Transformers, Datasets), scikit-learn
Research focus: Agent usability, trust and user control, explainable AI, evidence traceability, developer experience.

SELECTED RESEARCH AND PROJECTS

LLM Agentic Evaluation Benchmark for Data Wrangling Tasks

Ongoing | Manuscript in preparation for submission to ACL 2027

- Built an end-to-end benchmark and evaluation harness for LLM agents performing exploratory data wrangling on noisy, multi-domain datasets.
- Developed multidimensional task-quality rubrics and evaluated LLM-as-a-judge methods, achieving **90% agreement** with human assessments.
- Evaluated state-of-the-art LLMs and agents in sandboxed execution environments to assess their capabilities on open-ended data-wrangling tasks.

QualiVision | LLM Agents for Human-in-the-Loop Multimodal Analytics

Ongoing | VL/HCC 2026 Graduate Consortium

- Built a multimodal workspace linking video, transcripts, annotations, and analytic records; designing a synchronized data model for evidence traceability.
- Exploring LLM-agent workflows for multi-step pattern discovery and comparison, with researcher review and source evidence for verification.

How Wranglers Shape Wrangling | Human-AI Workflow Research

VL/HCC 2026

- Conducted a **40-participant** mixed-methods study of ChatGPT, Jupyter, Excel, and OpenRefine, assessing cleaning quality, workload, and user strategies.
- Synthesized performance, screen, and think-aloud evidence into recommendations for task-oriented guidance, visible data transformations, and human verification of AI assistance.

Reading Between the Curly Braces | Developer Experience Research

VL/HCC 2026

- Conducted a **215-participant** usability study and a **9-practitioner** interview to investigate reading and editing workflows across textual data formats.
- Found no consistent performance advantage across formats; translated evidence into design priorities around tooling, documentation, task complexity, and ecosystem support.

ChARLES | Explainable Analytics Interfaces

SIGMOD/PODS 2025 Companion

- Developed an interface for exploring model-generated explanations of database changes, with attribute recommendations, data-coverage views, and adjustable explanation complexity.

INDUSTRY EXPERIENCE

Data Scientist Intern | Lexical Intelligence, Rockville, MD | May - Aug 2022

Data Analytics Intern | Ermenegildo Zegna, Shanghai, China | Jan - Mar 2019